

# A cache-reuse trick cuts long-context serving cost, and it is a config flag.

Shared request prefixes get cached once across requests instead of being recomputed for every single one.

SERVING, INFERENCE & COST

EXAMPLE — CI VERIFICATION PENDING

## THE DIFF

- every request recomputes the shared prefix
- + the prefix is cached once, reused by all requests

### WHY IT MATTERS

Serving cost stops scaling with total tokens and starts scaling with unique tokens.

### WHO SHOULD CARE

Anyone serving one long system prompt, RAG preamble, or few-shot block to many users.

## THE NUMBER

# 65.7%

fewer tokens billed in the example below

- Savings scale linearly with how much of the request is repeated prefix.
- With  $n$  requests sharing a prefix, the ceiling on prefix cost is  $(n-1)/n$  of the prefix tokens.
- Below roughly 30% shared prefix the effect is not worth the config change.
- Not a quality change: outputs are identical, only the accounting differs.

## RUN IT — 9 lines, CPU, offline, <1s

```
def approx_tokens(text):
    return max(1, round(len(text) / 4))

prefix = "You are a helpful assistant. " * 40
requests = [prefix + q for q in ("hi", "why is the sky blue?", "can you say that again?")]

naive = sum(approx_tokens(r) for r in requests)
shared = approx_tokens(prefix) + sum(approx_tokens(r[len(prefix):]) for r in requests)

print(naive, shared, round(1 - shared / naive, 3))
```

881 302 0.657

## WATCH OUT

### CAVEAT

The prefix must be byte-identical -- one changed timestamp invalidates the cache. Cache residency is bounded by memory. A cache shared across tenants is a data-isolation decision, not just a performance one.

## SOURCES & LINKS — every claim above traces to one of these

| TYPE             | WHAT                            | URL                                                                                                                                                                                       |
|------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary source   | Original announcement / docs    | <a href="https://github.com/vllm-project/vllm">https://github.com/vllm-project/vllm</a>                                                                                                   |
| Runnable example | This exact script, tested in CI | <a href="https://github.com/marcocm28/ai-daily-diff/tree/main/examples/2026-08-28-prefix-cache">https://github.com/marcocm28/ai-daily-diff/tree/main/examples/2026-08-28-prefix-cache</a> |
| Slides           | All slides from the video, PDF  | <a href="https://marcocm28.github.io/ai-daily-diff/2026-08-28/slides.pdf">https://marcocm28.github.io/ai-daily-diff/2026-08-28/slides.pdf</a>                                             |